MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-736(*)/ASQ-88
AND
DECODER, PULSE KY-531(*)/ASQ-88

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide bearing, distance, and tone identity information to indicate the location of a complementary surface station with respect to the aircraft. As a discrete auxiliary function, the equipment shall also provide information to indicate the link distance with respect to a similarly equipped aircraft. This equipment is the airborne portion of the short-range air-navigation system known as TACAN (tactical air navigation.)

1.2 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (See paragraph 6.7).

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.9, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

FSC 5826

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3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials requirements shall be in accordance with Specification MIL-I-81036.

3.3 Design and Construction.

3.3.1 General - This equipment shall conform with all the applicable requirements of Specification MIL-I-81036 for design, construction and workmanship, except in the paragraphs as modified below.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance for 200 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.3.3.1 Cables and Connectors - Connectors used for interconnection of removable modular subassemblies shall be type D manufactured by Cannon Electric Company, or type 143 manufactured by Amphenol Electronics Corporation, or approved equivalents. Printed card connectors may be used in modules for mating chassis connectors where applicable. R. F. connectors used for interconnecting subassemblies within the r-f module shall be Sealectro type 3000, 3003, 3008, and 3012 or approved equivalents. UG-1465/U and UG-1464/U r-f connectors on subassemblies within the r-f module shall be Automatic Metal Products type 311-81450 (modified) and RF-02321 or approved equivalents. Internal r-f connectors used for connecting the r-f module to the chassis shall be Dage Electric type 30147-1 and 9722-1 or approved equivalents. External connectors shall be Amphenol Types 126-10572, 126-10753 or approved equivalents. The wire and terminals of the type 126-10753 connector shall be potted with a sealing compound which is capable of withstanding temperatures to +125°C.

3.3.6 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Military Specification MIL-I-6181B and MIL-I-81036 with the following changes:

3.3.6.1 Conducted Interference (Power Lead) - Power lead conducted interference shall be controlled within the limits of Specification MIL-I-6181B, except as follows:

Frequency Range	Level
0.15 to 0.25 mc	3000 uv/kc
0.25 to 20 mc	1000 uv/kc and 1000 uv cw

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3.3.6.2 Radiated Interference - Radiated interference shall be controlled within the limits of Specification MIL-I-6181B, except at the frequencies listed below. Radiated interference shall not exceed the indicated levels:

Frequency Range	Level
0.15 to 0.25 mc	30 uv/kc
20 to 50 mc	1 uv/kc
1/3 transmit frequency	200 uv cw
1/2 transmit frequency	200 uv cw
2/3 transmit frequency	200 uv cw

3.3.6.3 Antenna Conducted Interference - Antenna conducted interference shall be controlled within the following limits:

(1) Antenna conducted interference in the frequency range 0.15 to 960 mc shall be controlled within the limits of Specification MIL-I-6181B except at a frequency corresponding to one-third of the transmit frequency. The level at this frequency shall not exceed 500 microvolts.

(2) The antenna conducted CW power from the equipment, between transmitted pulses or in the receive mode of operation shall not exceed the following levels below the peak power of the transmitted pulses:

Frequency Range	Level
1015-1045 mc	100 db down
1046-1150 mc	80 db down

(3) At frequencies between 1250 and 10,000 mc, the antenna conducted output of the transmitter under transmit conditions shall not exceed a level of -30 dbw except that at frequencies corresponding to the third and fifth harmonics of the transmitter frequency, the antenna conducted interference shall not exceed a level of -10 dbw.

3.3.6.4 Receiver Interference Rejection - The equipment shall be capable of rejecting the following types of signals under the conditions specified:

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3.3.6.4.1 On Frequency - With the equipment tracking on a desired signal as indicated below, the specified performance shall not be affected by the following types and levels of interference at any frequency on or within 3 mc of the desired frequency, at the RT-736/ASQ-88 antenna connector:

Interference Type	Mode	Interference Level	Level of desired signal
6000 randomly spaced single pulses per second each 3.5 us wide	REC	-105 dbw	-105 dbw
	T/R	-105 dbw	-105 dbw
	A/A	-105 dbw	- 95 dbw
CW-within 300 kc of tracked signal	All	-130 dbw	-105 dbw
CW - 300 kc to 3 mc from tracked signal	All	-120 dbw	-105 dbw

3.3.6.4.2 Off Frequency - With the equipment tracking on a signal of -105 dbw, the specified performance shall not be affected when cw signals of the following frequencies and amplitudes are applied to the antenna connector through an appropriate filter:

Frequency	Amplitude
0.1 to 950 mc	-60 dbw
63 mc (IF)	-40 dbw
Image Frequency	-80 dbw
1250 to 2000 mc	-60 dbw
2000 to 3000 mc	-80 dbw
3000 to 4000 mc	-70 dbw
4000 to 7000 mc	-80 dbw
7000 to 10,000 mc	-60 dbw

3.3.9 Standard Conditions - In addition to the requirements of Specification MIL-I-81036, the following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests. All standard condition tests may be preceded by normal adjustment procedures.

Cooling Air:
(1) RT-736/ASQ-88 27.6 lb/hr of ambient room air
(2) KY-531/ASQ-88 41.8 lb /hr of ambient air

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B+ and Bias Voltages:	250+7V DC, 140 ± 2 VDC -180 V DC, as furnished from -265 V ±5V DC source through external series resistor of 4000± 10% ohms
Transmitter Load:	50 ohms resistive

3.3.10.1 Minimum External Cooling Requirements - The equipment shall operate satisfactorily under the temperature-altitude conditions specified for Class 2 equipment per Specification MIL-E-5400 when external cooling air is supplied at the rate indicated in Figures 1 & 2. The curves of Figures 1 & 2 specify the minimum cooling air flow rates that are required for air inlet temperatures varying from -40°F to +120°F. The cooling air weight flow as a function of the RT-736/ASQ-88 and the KY-531/ASQ-88 equipment pressure drops is specified in the following equations:

RT-736/ASQ-88

$$W = \left[.373 \quad (\sigma \Delta \rho)^{0.66}\right] \pm 10\% \text{ lb/min.}$$

KY-531/ASQ-88

$$W = \left[.578 \quad (\sigma \Delta \rho)^{0.53}\right] \pm 10\% \text{ lb/min.}$$

Where

W = Cooling Air flow rate (lb/min.)

$\sigma \Delta \rho$ = Corrected total pressure (inches H_2O)

σ = Density correction (ρ/ρ_r)

ρ_r = Standard density (0.0765 lb/ft^3)

3.3.12 Power Requirements - The equipment shall give specified performance with the power sources listed in Specification MIL-I-81036. The power requirements shall not exceed the specified amounts.

(1)	A-C Power, Phase 2 (to neutral)	
	26-Volts A-C	1.5 amperes
	115-Volts A-C	2.2 amperes
(2)	A-C Power, Phases 1 and 3 (to neutral)	
	115-Volts A-C	0.120 ampere per phase
(3)	D.C Power 28-Volt System	3.5 amperes, intermittent 0.5 ampere, continuous

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(4) D-C Power (From AM-2349/ASQ-19)

(a) +250V

Operating Voltage Limits	221 to 274 volts
Current	0.050 ampere at 250 volts

(b) +140V

Operating Voltage Limits	137 to 143 volts
Current	0.180 ampere at 140 volts

(c) -180V

Operating Voltage Limits	-238 to -300 volts source through external series resistor of 4000 ± 10% ohms
Current	0.024 ampere when source is -265 volts

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and service conditions. When reduced performance under extreme service conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified.

3.4.1 Operation - Receiver-Transmitter, Radio RT-736/ASQ-88 and Decoder, Pulse KY-531/ASQ-88, when installed in an aircraft, shall transmit to, and receive from, a similarly equipped aircraft or a complementary navigation beacon, a combination of pulsed signals from which it shall derive and process the information necessary to provide the following functions:

(a) Continuously display the bearing to the beacon in the RECEIVE (REC) and TRANSMIT-RECEIVE (T/R) modes.

(b) Indicate the deviation of the aircraft from the selected beacon radial in the REC and T/R modes.

(c) Indicate the distance to the beacon in the T/R mode.

(d) Indicate the distance to similarly equipped aircraft in the Air-to-Air (A/A) mode.

(e) In the A/A mode, transpond to interrogations from a similarly equipped aircraft.

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(f) Identify, by aural indication, the beacon to which the equipment is channeled in the REC and T/R modes.

(g) Automatically select one of two associated antennas in the REC, T/R, and A/A modes.

(h) Provide a blanking pulse output and a blanking pulse trigger synchronous with the radio's own transmissions in the T/R and A/A modes for the purpose of suppressing other equipment operating in the same frequency band.

3.4.1.1 Characteristics of TACAN Signals - This equipment shall operate with pulse code and pulse amplitude type signals of a surface or air facility meeting the following requirements:

3.4.1.1.1 Surface TACAN Signals - In the REC and T/R modes, this equipment shall provide the specified performance when operating with surface facilities which meet the requirements of MIL-STD-291.

3.4.1.1.2 Bilateral Ranging TACAN Signals - In the A/A mode, this equipment shall provide transponding and bilateral ranging service when operating with signals having the characteristics noted in 3.3.6.3, 3.4.2, 3.4.3.1, 3.4.3.2, 3.4.3.4, 3.4.3.5, 3.4.5.2, and 3.4.5.3.

3.4.1.1.3 TACAN Test Signals - Simulated TACAN signals used for test purposes shall be in accordance with MIL-STD-287.

3.4.1.2 Traffic Capacity in A/A Mode - With the equipment operating in the A/A mode on a given channel, it shall be possible to transpond effectively to as many as 24 properly equipped interrogating aircraft, when 95% of the aircraft are locked-on and are in the track mode, and 5% are in the search mode, when their equipments are each set 63 channels from the given channel, provided that the interrogation signal strengths are within the range of signal levels specified in 3.4.5.1 and do not differ from each other by an amount previously determined by the setting of the AGC adjustment specified in 3.4.4.5. The equipment specified herein shall interrogate and lock-on to one of the other aircraft, but it shall not be required to determine the one to which it is locked.

3.4.2 Frequency Characteristics - One hundred twenty-six crystal-controlled frequency channels shall be provided at 1 mc spacing. In the T/R mode, channels 1 through 63 shall pair receiver frequencies 962 through 1,024 mc with transmitter frequencies 1,025 through 1,087 mc; channels 64 through 126 shall pair receiver frequencies 1,151 through 1,213 mc with transmitter frequencies 1,088 through 1,150 mc. In the REC and T/R modes, the receiver shall be tuned identically. In the A/A mode, channels 1 through 63 shall pair receiver frequencies 1,088 through 1,150 mc with transmitter frequencies 1,025 through 1,087 mc; channels 64 through 126 shall pair receiver frequencies 1,025 through 1,087 with transmitter frequencies 1,088 through 1,150 mc.

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3.4.2.1 Channel Selection Time - Channel selection time from one channel to any other channel shall not exceed 7 seconds under standard conditions or 9 seconds under extreme service conditions.

3.4.3 Transmitter Characteristics - The TACAN and A/A interrogation signals and A/A reply signals transmitted by this equipment shall meet the following requirements:

3.4.3.1 Modulation Characteristics - The output pulse envelope as detected by a linear detector shall have a rounded shape falling within the limits outlined below:

3.4.3.1.1 Pulse Rise Time - The time required to rise from 10 to 90 percent of its maximum voltage amplitude shall not exceed 3.0 microseconds.

3.4.3.1.2 Pulse Fall Time - The time required for the trailing edge of the pulse to fall from 90 percent to 10 percent of its maximum voltage amplitude shall not exceed 3.5 microseconds.

3.4.3.1.3 Pulse Top - The amplitude of each pulse between the 90% voltage amplitude points shall be flat to within ± 1 db.

3.4.3.1.4 Pulse Duration - The pulse duration measured between 50% voltage amplitude points shall be 3.5 ± 0.5 microseconds.

3.4.3.1.5 Pulse Coding - In the T/R mode, the pulse code spacing measured between 50% voltage amplitude points from the leading edge of the first interrogation pulse to the leading edge of the second interrogation pulse shall be 12 + 0.5 microseconds. This requirement shall also apply to the interrogation pulses in the A/A mode. The transmitted replies in the A/A mode shall be single pulses.

3.4.3.2 Interrogation Pulse Repetition Rate - When the equipment is operating in the T/R or A/A modes, the pulse repetition rate shall not exceed 150 pulse pairs per second (pp/s) in the search condition, and shall not exceed 30 pulse pairs per second (pp/s) in the track condition. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted interrogations between any of 100 equipments on the same channel for more than one second in one hour. A search repetition rate of 135 pp/s shall be avaided to preclude synchronization with the fine bearing information.

3.4.3.3 Power Output

3.4.3.3.1 Power Output in T/R Mode - The peak power of each pulse of an interrogation pulse pair transmitted during the search condition shall be not less than +30 dbw under standard conditions or +27 dbw under service conditions when the transmitter is terminated in a 50 ohm resistive load impedance; under these conditions, the two pulses in an interrogation pulse pair shall not differ in peak power level by more than 1 db. The transmitted power output shall not drop below 55% of the preceding levels when terminated by a load which presents a voltage standing wave ratio of 2.1 at any phase angle.

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3.4.3.3.2 Power Output in A/A Mode - With the equipment transmitting at the search interrogation rate, plus transponding with single pulse replies to the interrogations of one complementary aircraft as specified in 3.4.5.1, the requirements of 3.4.3.3.1 shall apply. When the equipment is transponding to the interrogations from up to 24 complementary aircraft, in accordance with paragraphs 3.4.1.2 and 3.4.5.2, the average peak pulse power shall be not less than +27 dbw under standard conditions or +24 dbw under service conditions, into a 50 ohm resistive load. The transmitter shall be capable of recovery to within 2 db of the specified average peak pulse power within 100 microseconds of a previous transmission.

3.4.3.4 Frequency Stability - The center frequency of the transmitted signal shall be within ± 100 kc of the specified channel frequency.

3.4.3.5 Transmitter Spectrum - In both the search and track conditions, 90% of the transmitted energy shall be within a 500 kc band centered on the specified frequency of that channel, and the remaining energy shall decay in an essentially smooth manner with increasing or decreasing frequency.

3.4.3.6 Transmitter Muting - The transmitter shall be automatically muted during channeling in the T/R and A/A modes.

3.4.4 Receiver Characteristics - The radio receiver of this equipment shall provide the following performance in the REC, T/R, or A/A modes, as applicable:

3.4.4.1 Sensitivity - The equipment sensitivity for tone identity, bearing, distance and transponding shall be as prescribed in 3.4.5.1, 3.4.6.1, 3.4.7.2, and 3.4.8.1 respectively.

3.4.4.2 Decoding - In the REC and T/R modes, the sensitivity of the receiver shall not decrease by more than 3 db when receiving a TACAN pulse pair spaced ± 0.5 microsecond from nominal X channel pulse coding. Loss of receiver sensitivity shall be at least 30 db for TACAN pulse pairs spaced ± 4.0 microseconds from nominal X channel pulse coding. The foregoing shall also apply to transponding sensitivity when in the A/A mode, for the transmission of A/A replies synchronous with the received A/A interrogations. In the A/A mode, the received signals shall be applied to the distance measuring circuity without decoding.

3.4.4.2.1 Suppression of Decoding - Decoding shall be suppressed during the period starting not more than two microseconds after the external suppression pulse described in 3.4.10.3 has reached 15 volts amplitude and lasting throughout the period the external suppression pulse remains at or above 15 volts amplitude. Means shall be provided to prevent decoding of this equipment's own interrogation transmissions. All related performance parameters of this specification apply only when there is no suppression of decoding by external equipment.

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3.4.4.3 Selectivity - The equipment shall have 50 db or greater rejection to any normal beacon signal or signals on channels other than the channel to which the equipment is tuned. This requirement applies both in the presence or absence of a desired signal. The undesired signal shall be considered rejected if the performance requirements of this specification are met, with the exception of allowing an additional 1-degree bearing error. In the absence of a desired signal, the undesired signal shall be considered rejected if no distance, bearing, or identity tone information is obtained.

3.4.4.4 Amplitude Discrimination - In the REC and T/R modes, the equipment shall be capable of rejecting the distance and bearing information from the weaker of two beacon signals received on the same channel when the difference in signal strength is 8 db or more. The performance obtained from the stronger signal shall meet the requirements of this specification, except that an additional 1-degree bearing error shall be permitted when the two signals differ in level by 8 db. The 8 db of difference shall be measured using test signals, each having an equal modulation percentage but not to exceed a total of 40% modulation. In the A/A mode, the equipment shall be capable of rejecting A/A interrogations when their signal levels are weaker than the strongest A/A interrogation by the amount specified in 3.4.4.5.

3.4.4.5 Automatic Gain Control (AGC) - The AGC signal shall be generated subsequent to decoding. In the air-to-air mode, an adjustable AGC circuit shall be provided. This AGC shall be adjustable from no AGC action to the normal AGC setting which will prevent decoding of any A/A interrogation pulse pair which has a signal strength 8 db or more lower than that of the strongest A/A interrogation received, provided that the strongest interrogation is not in excess of -30 dbw; this shall include the case where the equipment is channeled into an A/A interrogation which is operating at a track interrogation rate as low as 25 pp/s, and this interrogation is the strongest received. When an abrupt change is made in the A/A interrogation signal strength applied to the RT-736/KY-531 in the air-to-air mode, the AGC shall reach 90% of its final value in less than 5 seconds. Unless otherwise specified, all related performance requirements apply only with the normal A/A AGC setting.

3.4.5 Transponding Characteristics - In the A/A mode this equipment shall transpond to received, properly coded, A/A interrogations with single reply pulses in accordance with the following requirements:

3.4.5.1 Triggering Level - The equipment shall transpond with the reply efficiency specified in paragraph 3.4.5.2, when A/A interrogations from a similarly equipped aircraft only, are received at a signal level (at the RT-736 antenna connector) as indicated below:

Conditions	Signal Level
Standard	-114 dbw to -30 dbw
Service	-111 dbw to -30 dbw

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3.4.5.2 Reply Efficiency - Under the conditions noted in 3.4.5.1, the average reply efficiency shall be at least 90%. When the equipment is interrogated by as many as 24 similarly equipped aircraft, in accordance with 3.4.1.2, assuming a random distribution of A/A interrogations, the average transponding reply efficiency shall be at least 70%, provided that this equipment is locked-on to one of the other aircraft.

3.4.5.3 Transponder Delay - The delay at the RT-736/ASQ-88 antenna connector between receipt of the second pulse of an A/A interrogation pulse pair and the transmission of the corresponding single A/A reply pulse shall be as follows:

Interrogation Level	System Delay
6 db above minimum triggering level to -100 dbw	$50 + 4^{-2}$ microseconds
-99 dbw to -30 dbw	50 ± 1 microseconds

All delay times shall be measured between the 50% voltage amplitude points on the leading edges of the respective pulses.

3.4.5.4 Interrogation Priority - Means shall be provided to give the A/A interrogations transmitted by this equipment priority over the transmission of replies to A/A interrogations from similarly equipped aircraft. The transmission of A/A replies shall be inhibited during the period from at least 150 microseconds preceeding to at least 50 microseconds following the transmission of the first pulse of an A/A interrogation pulse pair by this equipment.

3.4.5.5 Automatic Overload Control (AOC) - An AOC circuit shall be provided to protect the transmitter from damage due to overinterrogation. This circuit shall operate to reduce the receiver gain after the threshold transmitter prf has been reached. The AOC threshold shall be adjustable to the point where the requirements of 3.4.5.2 are met, while holding the transmitter duty cycle within safe limits.

3.4.5.6 Random Triggering Rate - In the A/A mode, the equipment shall not transmit more than 20 A/A reply single pulses per second, as averaged over a one minute period, under the following conditions:

(1) The RT-736/ASQ-88 antenna connector is terminated in a 50 ohm resistive load.

(2) No external signal is present.

(3) The interrogation generator is disabled.

3.4.6 Bearing Measuring Characteristics - In the REC and T/R modes, this equipment shall meet the following bearing measuring requirements. In the STANDBY (STBY) and A/A modes, means shall be provided to disable the bearing servo.

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3.4.6.1 Bearing Sensitivity and Accuracy - With nominal signal modulation, the bearing sensitivity and accuracy as measured with precision test equipment at the output of the fundamental synchro resolver and the synchro differential transmitter (with a simulated zero degree signal applied to the compass input) shall be as follows:

STANDARD CONDITIONS

Bearing	Bearing Accuracy	Signal Level
270 degrees	± 2 degrees	Not greater than -117 dbw
All bearings	± 2 degrees	Not greater than -113 dbw
All bearings	± 1 degree	-107 dbw to -35 dbw
All bearings	± 2 degrees	-35 dbw to -30 dbw

SERVICE CONDITIONS

Bearing	Bearing Accuracy	Signal Level
270 degrees	± 2.5 degrees	Not greater than -115 dbw
All bearings	± 2.5 degrees	Not greater than -110 dbw
All bearings	± 1.5 degrees	-104 dbw to -35 dbw
All bearings	± 2.5 degrees	-35 dbw to -30 dbw

3.4.6.2 Bearing Tracking Rate - The bearing system shall lock-on and track signals having the bearing changing at rates up to 20 degrees per second in either direction for signal levels from -107 dbw to -30 dbw under standard conditions and -104 dbw to -30 dbw under service conditions, except that at ambient temperatures below -30°C, the tracking rate shall be up to 10 degrees per second. After an additional 15 minutes of operation, the bearing system shall lock-on and track signals having bearings changing at rates up to 20 degrees per second.

3.4.6.3 Bearing Search Time - A complete bearing search cycle shall not exceed 18 seconds with or without a signal applied except that, at ambient temperatures below -30°C, a complete bearing search cycle shall not exceed 40 seconds after an intial 5 minutes warmup time. With an additional 15 minutes of operation, a complete bearing search cycle shall not exceed 18 seconds.

3.4.6.4 Bearing Memory - Upon loss of signal, the memory period during which the equipment retains its bearing reading shall fall between 8 and 12 seconds.

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3.4.6.5 Coarse Bearing System Phase Stability - The phase angle of the pulse produced from the 15 cps modulation with respect to the pulse produced from the north reference burst shall be held within ± 7.5 degrees of the correct value under standard conditions and within ± 10 degrees under service conditions. This requirement shall be met with signal modulation levels from 15 to 55%.

3.4.6.6 Bearing Instrumentation Output Characteristics - The equipment shall provide suitable bearing output circuits for operation with up to three ID-663/U indicators and one ID-387/ARN indicator with its deviation indicator terminals paralleled with a 500 ohm resistor or two standard deviation indicator loads. The bearing output circuits shall be in accordance with the following requirements:

3.4.6.6.1 Course Deviation Output - This output shall be presented to the vertical needle of the ID-387/ARN as a deviation from the manually selected course to-or from-the beacon. The impedance into which it operates shall be maintained at 333 ohms. The course deviation sensitivity shall be adjustable so that a course error of 5 degrees will provide 450 microamperes into the specified load impedance.

3.4.6.6.2 To-From Circuit Output - The to-from circuit shall be capable of supplying the ambiguity indicator of the ID-387/ARN with d-c of either polarity and a minimum magnitude of 250 microamperes for ± 20 degrees course error. Polarity shall be determined by the relative phase of the 400-cps signal input to the to-from circuit.

3.4.6.6.3 Bearing Flag Alarm Output - The bearing flag alarm circuit shall provide a terminal which furnishes +28V dc to the ID-387/ARN flag alarm when the bearing circuits are properly tracking a TACAN signal or are in memory, and which is open circuited when the bearing circuits are not properly tracking. This terminal shall also be open circuited in the STBY and A/A modes of operation.

3.4.6.6.4 Magnetic Bearing Output - A synchro resolver rated for 26-V 400-cps excitation and 12.6 volt line-to-line maximum output shall be provided for magnetic bearing electrical output. The resolver shall be set at 400 cps electrical zero (EZ) at 300 degrees indicated bearing. This output shall be used in conjunction with the manually selected course resolver in the ID-387/ARN to establish the deviation from the selected course.

3.4.6.6.5 Relative Bearing Output - This output shall be presented to the ID-663/U as combined magnetic-heading-bearing signal. A synchro of the differential control type shall be provided to combine an electrical compass heading signal from an Eclipse Pioneer AY-201-1, or equivalent, synchro transmitter so as to produce an electrical output suitable for actuating the synchro-receiver-driver relative bearing pointers of the ID-663/U. An output accuracy degraduation of ± 0.5 degree is permissible when the equipment is loaded with an impedance equivalent to three ID-663/U instruments in parallel.

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3.4.7 Distance Measuring Characteristics - In the T/R and A/A modes, as applicable, this equipment shall meet the following distance measuring characteristics. In the STBY and REC modes, means shall be provided to disable the distance servo.

3.4.7.1 Operational Range - The operational range of the distance measuring circuitry shall be from zero to 195 nautical miles. Under no circumstances shall the equipment lock on to its own transmission.

3.4.7.2 Distance Sensitivity and Accuracy - As measured at the output of the distance synchro transmitters, with precision test equipment, the equipment shall have the following distance-measuring performance when operating with a transponding facility which is 90% efficient in replying to interrogations:

Conditions	Accuracy	Signal Level
Standard	± (0.2 mile + 0.1%)	-117 dbw to -30 dbw
Service	± (0.3 mile + 0.1%)	-114 dbw to -30 dbw

3.4.7.3 Distance Tracking Rate - The equipment shall be capable of locking-on and continuously tracking distance replies under the conditions listed below. The speeds listed shall be the relative speed between this equipment and the transponding equipment, either inbound or outbound:

	Signal Level			
Mode	Standard Cond.	Service Cond.	Transponder Reply Efficiency	Speed (knots)
T/R	-111 dbw to -30 dbw	-108 dbw to -30 dbw	70%	2500
A/A	-111 dbw to -30 dbw	-108 dbw to -30 dbw	85%	3000

3.4.7.4 Distance Search Speed - The distance search speed shall average at least 6.9 miles per second, with or without a signal applied. The distance search shall be in an outbound direction (progressively increasing range).

A complete distance search cycle shall not exceed 30 seconds when the transponder reply efficiency is at least 70% during the period that the ranging circuits intercept the reply signal.

3.4.7.5 Distance Memory - Upon loss of signal, the memory period during which the equipment retains its distance reading shall fall between 4 and 8 seconds.

3.4.7.6 Distance Zero Calibration - During bench adjustments preceeding standard condition tests, the equipment may be adjusted for an optimium distribution of positive and negative distance errors; however the calibration shall be such that the distance error at zero miles range is in accordance with the following requirements:

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3.4.7.6.1 Calibration in A/A Mode - Under standard conditions in the A/A mode the equipment shall be adjusted to read zero miles ± 0.1 mile, as measured at the output of the units synchro transmitter with precision test equipment, when the delay at the antenna terminal between the 50% voltage amplitude points on the respective leading edges of the second A/A r-f interrogation pulse of a coded transmitted pair and the corresponding received r-f reply pulse is 50 microseconds.

3.4.7.6.2 Calibration in T/R Mode - Under standard conditions in the T/R mode the distance output shall indicate zero miles within the accuracy specified in 3.4.7.2, when the delay at the antenna terminal between the 50% voltage amplitude points on the respective leading edges of the second T/R r-f interrogation pulse of a coded transmitted pair and the corresponding second r-f reply pulse is 50 microseconds.

3.4.7.7 Distance Instrumentation Output Characteristics - This equipment shall provide suitable output circuits for operation with the distance indicators of up to three ID-663/U instruments, in accordance with the following requirements.

3.4.7.7.1 Units and Tens Outputs - Synchro transmitters, rated at 26-volt 400-cps input, shall be provided to produce electrical distance output signals to the synchro torque receivers driving the units and tens dials of the ID-663/U. An output accuracy degradation of ± 0.03 miles in the units output is permissable when the equipment is loaded with an impedance equivalent to three ID-663/U instruments in parallel.

3.4.7.7.2 Hundreds Output - A switch shall be provided, actuated by the distance gear train, for the purpose of controlling the hundreds dial of the ID-663/U. When the measured distance is between 0 and 99 miles, 400-cps of the hundreds dial of the ID-663/U. When the measured distance is between 100 and 195 miles, the switch shall be actuated so as to cause an indication of one on the hundreds dial of the ID-663/U.

3.4.7.7.3 Distance Flag Alarm - The distance flag alarm circuit shall provide a terminal which is grounded when the distance circuits are properly tracking a TACAN signal or are in memory, and which is ungrounded when the distance circuits are not properly tracking. A terminal shall also be provided to furnish +28v dc to the distance flag alarm in the ID-663/U.

3.4.8 Identification Characteristics - In the REC and T/R modes, the equipment shall provide for aural identification of the beacon to which the equipment is channeled by a Morse coded audio tone of 1350 cps in accordance with the following requirements. Unless otherwise noted, these requirements apply only to beacon identification signals in accordance with MIL-STD-291.

3.4.8.1 Sensitivity - The equipment shall provide intelligible and unambiguous identification output with a signal input from -117 dbw to -30 dbw under standard conditions, and -114 dbw to -30 dbw under service conditions. For beacons without equalizing pulses, the identity sensitivity shall be -111 dbw or better.

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3.4.8.2 Identity Output - At signal levels 3 db or more above the minimum levels specified in 3.4.8.1, the signal-to-noise ratio (measured with tone and squitter signals applied in sequence) shall be not less than 30 db and the audio output shall be not less than 50 milliwatts nor more than 500 milliwatts as measured with a 600-ohm load; for beacons without equalizing pulses, the audio output shall be at least 5 milliwatts and the signal-to-noise ratio shall be at least 10 db. Accidental shorting or opening of the external load shall not cause damage to the equipment or render other functions inoperative.

3.4.9 Antenna Selection - This equipment shall be capable of operating with an external antenna switching control unit which switches the RT-736/ASQ-88 antenna connector alternately between each of two antennas in such a manner that the transfer of antennas is affected in not less than two seconds nor more than four seconds after the start of a sampling period for a given antenna, until a "switching inhibit" command signal is supplied by the KY-531/ASQ-88. If the "switching inhibit" command signal is supplied not later than 75 milliseconds before the end of a sampling period for a particular antenna, that antenna shall remain connected to the RT-736/AS-99 antenna connector as long as the "switching inhibit" command signal remains above the minimum amplitude specified in paragraph 3.4.9.1. When the "switching inhibit" command signal ceases to be present, antenna switching action shall resume in not more than four seconds.

3.4.9.1 Antenna "Switching-Inhibit" Command Signal Characteristics - The "switching inhibit" command signal output from the KY-531/ASQ-88 shall have the following characteristics into a 5600 ohm load resistance:

A. Command Present

(1) Amplitude - 0.5 to 20 volts RMS

(2) Waveshape - The waveshape shall be similar to that produced by positive-going half-wave rectification of the 380-420 cps power source.

B. Command Absent -

(1) Residual Noise - 200 millivolts, RMS, maximum.

The command signal shall be generated when the decoded signals received by the RT-736/ASQ-88 and KY-531/ASQ-88 meet the following requirements.

Mode	Average PRF of Decoded Pulse Pairs
REC	700 pps or greater
T/R	700 pps or greater
A/A	25 pps or greater

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In the A/A mode, momentary dropouts of the command signal are acceptable during gaps in the decoded information exceeding 50 milliseconds due to countdown and/or excessive jitter in the received interrogation signal.

3.4.10 Suppression Pulse Characteristics - This equipment shall generate a blanking pulse and a blanking pulse trigger as described below, synchonous with its own transmissions, for the purpose of initiating suppression of other equipment operating in the same frequency band. This equipment shall also accept blanking pulses, as described below, from external equipment for the purpose of suppression of the decoding of received signals in accordance with 3.4.4.2.1.

3.4.10.1 Blanking Pulse Trigger Output - This equipment shall generate a single blanking pulse trigger, with characteristics specified below, for each interrogation pulse pair or each A/A reply transmitted. This trigger shall normally be used in the AN/ASQ-88 system to initiate suppression of the IFF receiver.

(1) Polarity: Positive

(2) Amplitude: 15 to 70 volts

(3) Load Impedance: 2200 ohms in parallel with 500 picofarads at the blanking pulse trigger output terminal.

(4) Rise Time: 0.7 microsecond maximum from 3 volt to 10 volt level

(5) Duration: 1.5 to 20 microseconds at 10 volt level; duration at the 3 volt level shall not exceed 30 microseconds.

(6) Synchronization: The 10 volt level on the leading edge shall occur at least 0.5 but not more than 5.0 microseconds before the point at which the transmitted A/A reply pulse or the first pulse of the transmitted interrogation pulse pair reached 10% of its maximum voltage amplitude.

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(7) Spurious Output: Any spurious output at the blanking pulse trigger terminal shall not exceed ± 1.0 volt in amplitude with the specified load impedance. Spurious output shall be defined as any output outside of the maximum dimensions of the blanking pulse trigger envelope permitted above, with an additional 30 microseconds allowed for the decay from the 3 volt to the 1 volt level.

(8) Operating Mode: Blanking pulse triggers shall be generated only when the equipment is in the T/R or A/A modes of operation.

3.4.10.2 ELRAC Blanking Pulse Output - The equipment shall generate a single blanking pulse, with characteristics specified below, for each interrogation pulse pair or each A/A reply transmitted. This blanking pulse shall be used for suppression of ELRAC equipment.

(1) Polarity: Positive

(2) Amplitude: 5 ± 2 volts

(3) Load Impedance: 93 ohms, resistive

(4) Rise Time: 0.5 microsecond maximum from the 1 volt to the 3 volt level.

(5) Duration: 26 ± 3 microseconds at the 3 volt level. Duration at the 1 volt level shall not exceed 30.5 microseconds.

(6) Synchronization: The blanking pulse shall reach 3 volts amplitude 0.5 to 5.0 microseconds ahead of the point at which the transmitted A/A reply pulse or the first pulse of a transmitted interrogation pair reaches 10% of its maximum voltage amplitude.

(7) Spurious Output: Any spurious output at the ELRAC blanking pulse connector shall not exceed ± 0.5 volt in amplitude with the specified load impedance. Spurious output shall be defined as any output outside of the maximum dimensions of the ELRAC blanking pulse envelope permitted above, with an additional 30 microseconds allowed for the decay from the 1 volt to the 0.5 volt level.

(8) Operating Mode: Blanking pulses shall be generated only when the equipment is in the T/R or A/A modes of operation.

3.4.10.3 Suppression from External Source - An input shall be provided to receive an externally generated blanking pulse. This pulse shall desensitize the receiving function as described in 3.4.4.2.1.

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In the AN/ASQ-88 system this blanking pulse will normally be supplied by the IFF equipment. The suppression input shall present the following load impedance and operate with blanking pulses having the following characteristics:

(1) Polarity: Positive

(2) Amplitude: 15 to 30 volts

(3) Load Impedance: Not less than 4700 ohms in parallel with 100 picofarads.

(4) Rise Time: 1.0 microseconds maximum from the 3 volt to the 15 volt level.

(5) Synchronization: The blanking pulse shall reach 15 volts amplitude within the period 5.0 microseconds preceding to 3.0 microseconds following the point at which the first transmitted pulse of a train reaches 10% of its maximum voltage amplitude.

(6) Duration: The blanking pulse shall remain at 15 volts or greater amplitude from 7 to 17 microseconds after the trailing edge of the last pulse of the transmitted pulse train has decayed to 10% of its maximum voltage amplitude.

(7) Decay Time: 2.0 microseconds maximum from the 15 volt to the 3 volt level.

(8) Spurious Output: Neither the DC steady-state output potential nor the instantaneous peak AC potential of spurious signals shall exceed ± 1.0 volt.

3.5 Detail Requirements

3.5.1 Receiver-Transmitter, Radio RT-736/ASQ-88 - The radio receiver-transmitter shall meet the following requirements.

3.5.1.1 Function - When installed in an AN/ ASQ-88, the radio receiver-transmitter shall perform the following functions:

(1) In response to trigger signals supplied from Decoder, Pulse KY-531/ASQ-88, generate pulsed r-f signals on the selected frequency channel for interrogation of TACAN navigation beacons in the T/R mode, other similarly equipped aircraft in the A/A mode and replies to similarly equipped aircraft interrogations generated in accordance with 3.4.10.1

(2) Receive and demodulate the TACAN beacon bearing signals, distance replies, and identification as applicable in the REC and T/R modes, and interrogation and reply signals from similarly equipped aircraft in the A/A mode.

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(3) Provide a video output signal, at a level suitable for operation of Decoder, Pulse KY-531/ASQ-88, containing the composite demodulated bearing signals, distance replies and identification signals as applicable in the REC and T/R modes, and interrogation and reply signals in the A/A mode.

(4) Provide a video signal for operation of the deblocker circuitry in the KY-531/ASQ-88.

3.5.1.2 Form Factor - The radio receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shock-mounted equipment shelf which in turn is mounted in an aircraft. The over-all dimensions, excluding protrusions, shall not exceed 7-17/32 by 7-37/64 by 22-37/64 inches. The front panel shall extend an additional 1-1/32 inches below the base. Protrusions of 17/64 inches shall be allowed on each side in addition to the above dimensions.

3.5.1.3 Weight - The weight of the receiver-transmitter shall not exceed 40.0 pounds.

3.5.1.4 Contents - The radio receiver-transmitter shall include a multifrequency crystal-controlled receiver and transmitter, modulator, power supply, and other circuitry required for performance of the specified functions. The unit shall be contained in a pressurized case and shall include a heat exchanger and internal blower as necessary for cooling. The following major sub-assemblies are to be included:

R-f module

Power Supply Module

Base chassis, with internal blower, heat exchanger, filament transformer,

fuses, elapsed-time meter, connectors, wiring, and associated cover.

3.5.1.5 Frequency Measuring Facilities - A coaxial test jack will be contained in the r-f module to permit a convenient sampling point for frequency measurements. This jack shall be coupled to the 90 mc circuitry in the frequency multiplier chain. It shall be possible to determine that all transmitter frequency generating elements are functioning by measuring the frequency of not more than 13 channels.

3.5.1.6 Channeling Mechanism - The receiver-transmitter shall be automatically tuned to the desired channel as selected by the appropriate switches of Control, Central C-6248/ASQ-88. The channeling mechanism shall be activated only during the tuning cycle, and mechanically latched in place when on the desired channel.

3.5.1.7 Preselector - The preselector shall cover the complete tuning range from 962 to 1213 mc. The use of solenoid-actuated L-Band switching devices in the preselector shall not be permitted.

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3.5.1.8 Solid-State Circuitry- The RT-736/ASQ-88 shall utilize solid-state circuitry to the maximum practicable extent, consistent with other requirements. In this unit, vacuum tubes shall be used only in R-f circuitry which operates above 340 mcs. The use of electromechanical relay-type devices shall be kept to a minimum.

3.5.1.9 Pressurization - The RT-736/ASQ-88 shall be pressurized to permit operation in accordance with this specification at 70,000 feet. This unit shall be equipped with a pressure-relief valve and a make-up air valve to be used in conjunction with the make-up air system in the aircraft. The equipment performance shall be in accordance with this specification when the internal pressure is at 20 ± 5 psia.

3.5.1.10 Controls

Location	Function
Power Supply Module	+10 volt adjust
Power Supply Module	-18 volt adjust
Power Supply Module	time delay adjust
Power Supply Module	low voltage adjust
Modulator - R-F Module	12 microsecond pulse spacing adjust
Channel Selector R-F Module	IF gain adjust

3.5.1.11 Electrical Connections - Connections to external circuits shall be provided as follows:

Receptacle Type	Function
*Industrial Products 81450 (or approved equivalent)	antenna connector
Amphenol 126-10572 (or approved equivalent)	power and control connections

*This connector shall be incorporated as part of an antenna filter assembly.

3.5.1.12 External Finish - The equipment covered by this specification shall be painted with heat-reflective aluminum paint.

3.5.1.13 Antenna Impedance - The equipment shall operate with antennas and coaxial r-f transmission lines of 50 ohms impedance. The equipment shall not be damaged by operation in the transmit-receive or the air-to-air mode with the antenna connector short-circuited or open-circuited.

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3.5.2 Decoder, Pulse KY-531/ASQ-88 - The pulse decoder shall meet the following requirements:

3.5.2.1 Function - When installed in the AN/ASQ-88, the pulse decoder shall perform the following functions:

(1) Supply to Receiver-Transmitter, (Radio RT-736/ASQ-88) trigger signals for initiating the r-f interrogation pulses and r-f A/A reply pulses.

(2) Accept the composite video signal from Receiver-Transmitter, (Radio RT-736/ASQ-88) and use this signal to determine distance, bearing and identification relative to the TACAN navigation beacon in the T/R mode. In the A/A mode, the decoder will use the composite video signal to determine distance relative to a complementary aircraft and to generate appropriate transponding reply trigger pulses.

(3) Accept a video signal from the RT-736/ASQ-88 for operation of the deblocker circuitry.

(4) Provide an output for control of receiver gain in the RT-736/ASQ-88. This output shall contain composite AGC and AOC information, as required by received signal strength and operating mode.

(5) Provide a synchro resolver output which will supply Indicator, Course ID-387/ARN, magnetic bearing information.

(6) Provide suitable output signals for the operation of the deviation indicator and to-from indicator of the ID-387/ARN.

(7) When supplied a magnetic heading signal from the aircraft compass system, provide a synchro output suitable for actuating a bearing pointer of the ID-663/U indicator.

(8) Supply sunchro signals for actuating the distance dials of the ID-663/U.

(9) Provide facilities for the actuation of flag alarm circuits of the ID-387/ARN and ID-663/U indicators.

(10) Provide an audio frequency output signal suitable for aural identification of the TACAN navigation beacon in the REC and T/R modes.

(11) Provide a blanking pulse output suitably synchronized with transmitted interrogations for suppression of ELRAC equipment during transmit periods.

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(12) Provide a switching-inhibit signal for use with external antenna switching equipment.

3.5.2.2 Form Factor - The pulse decoder shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which is in turn mounted in an aircraft. The over-all dimensions, excluding protrusions, shall not exceed 7-9/16 by 6-15/32 by 22-33/64 inches. The front plate shall extend an additional 1-1/32 inches below the base of the unit. Protrusions of 5/64 inches shall be allowed on each side in addition to the above dimensions.

3.5.2.3 Weight - The weight of the pulse decoder unit shall not exceed 35.5 pounds.

3.5.2.4 Contents of Pulse Decoder KY-531/ASQ-88 - The pulse decoder unit shall contain all circuitry and mechanisms necessary to perform the required functions. The following major subassemblies are to be included:

Interrogation Generator Module

Distance Servo Amplifier Module

Time Discriminator Module

Distance Decoder Module

Fine Bearing Detector Module

Coarse Bearing Detector Module

Bearing Decoder Module

Bearing Servo Amplifier Module

Bearing Unit (Coupler)

Distance Unit (Coupler)

Course Converter (Phase Detector)

Relay Unit

Electronic Control (AGC/AOC Amplifier)

Chassis, with connectors, wiring, power-supply subassembly, fuses, elapsed-time meter, relays and a subassembly containing A/A and ELRAC circuitry

Top and Bottom Covers

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3.5.2.5 Controls

Location	Function
Fine Bearing Detector Module	Track Adjust
Fine Bearing Detector Module	Zero Adjust
Coarse Bearing Detector Module	15 cps Gate Centering Adjust
Bearing Decoder Module	North Reference Gate Width Adjust
Distance Decoder Module	Aperture Width Adjust
Time Discriminator Module	Sweep Tracking Adjust
Interrogation Generator Module	Oscillator Level Adjust
Interrogation Generator Module	Zero Adjust
Interrogation Generator Module	Resolver Phase Adjust
Interrogation Generator Module	RATE Adjust
Course Converter	Deviation Centering Adjust
Course Converter	Deviation Gain Adjust
AGC/AOC Electronic Control Amplifier	AUTOMATIC Reset Adjust
AGC/AOC Electronic Control Amplifier	T/R Video Level Adjust
AGC/AOC Electronic Control Amplifier	A/A Video Level Adjust
AGC/AOC Electronic Control Amplifier	AOC Adjust
A/A Subassembly	A/A Transponding Delay Adjust
Power Supply Subassembly	-18V Adjust
Power Supply Subassembly	+10V Adjust

3.5.2.6 Electrical Connectors - Connections to external Circuits shall be provided as follows;

Receptacle Type	Function
Amphenol 126-10753 (or an approved equivalent)	Power and Control Connections

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3.5.2.7 External Finish - The equipment covered by this specification shall be finished with heat-reflective aluminum paint. The bottom cover of the equipment shall not be painted.

3.5.2.8 Test Panel - Test points shall be provided on the front of the KY-531/ASQ-88 for use with suitable test equipment. The following voltages shall be provided to the test points:

Test Point Designation	Function
J22	Video
J23	AGC
J24	26 V AC
J25	FIL.
J26	+250 VDC
J27	+140 VDC
J28	+36 VDC
J29	+28 VDC
J30	+10 VDC
J31	-18 VDC
J32	-40 VDC
J33	NORMAL Trigger (T/R)
J34	A/A Trigger
J35	Distance Reply
J36	Ant. Sw.

3.5.2.9 Output Requirements - The equipment shall provide suitable output circuits for operation with up to three ID-663/U indicators and one ID-387/ARN indicator with its deviation indicator terminals paralleled with a 500-ohm resistor.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the test requirements of MIL-I-81036 to determine compliance with all applicable requirements.

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4.1.1 Environmental Testing - All environmental testing shall be in accordance with the procedures of MIL-T-5422 during steps 2, 3, 4, 6, 7, 9 10, 11, 12, and 14 outlined in MIL-T-5422; cooling air shall be supplied to the RT-736/ASQ-88 and KY-531/ASQ-88; and 3-phase 115 VAC 400 cps power shall be supplied to the internal blower of the RT-736/ASQ-88. This shall also apply for at least 30 minutes prior to applying any other power to this equipment when operation is required during or following the humidity test specified in MIL-T-5422.

5. PREPARATION FOR DELIVERY

5.1 General - Preparation for delivery shall be in accordance with Specification MIL-I-81036.

6. NOTES

6.1 Use - The equipment covered by this specification is intended for aircraft installation as a portion of the Integrated Electronic Central, AN/ASQ-88, to provide TACAN navigation serivce, including A/A ranging.

6.2 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) Specification MIL-I-81036 shall have precedence over this specification

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with Specification MIL-I-81036. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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6.4 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.5 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.6 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 1.2).

6.8 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of this equipment:

Amplifier-Power Supply-Receiver AM-2349/ASQ-19

Earphone H-87A/U

Antenna AT-234/APX

Indicator, Course ID-387/ARN

Control, Central C-6248/ASQ-88

Intercommunication Station LS-460/AIC

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Indicator, Bearing-Distance Heading ID-663/U

6.9 General - The following documents of the issue in effect on the date of invitation for bids, except the dated documents indicated below, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181B (29 May 1953)	Interference Limits, Test and Design Requirements, Aircraft Electrical and Electronics Equipment
MIL-I-81036	Integrated Electronic Central AN/ASQ-88

STANDARDS

MILITARY

MIL-STD-287	TACAN Standard Test Signal
MIL-STD-291	TACAN Standard Signal

6.9.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Specification MIL-I-6181B is available only from Bureau of Naval Weapons, Washington 25, D. C., Attention RAAV-34.

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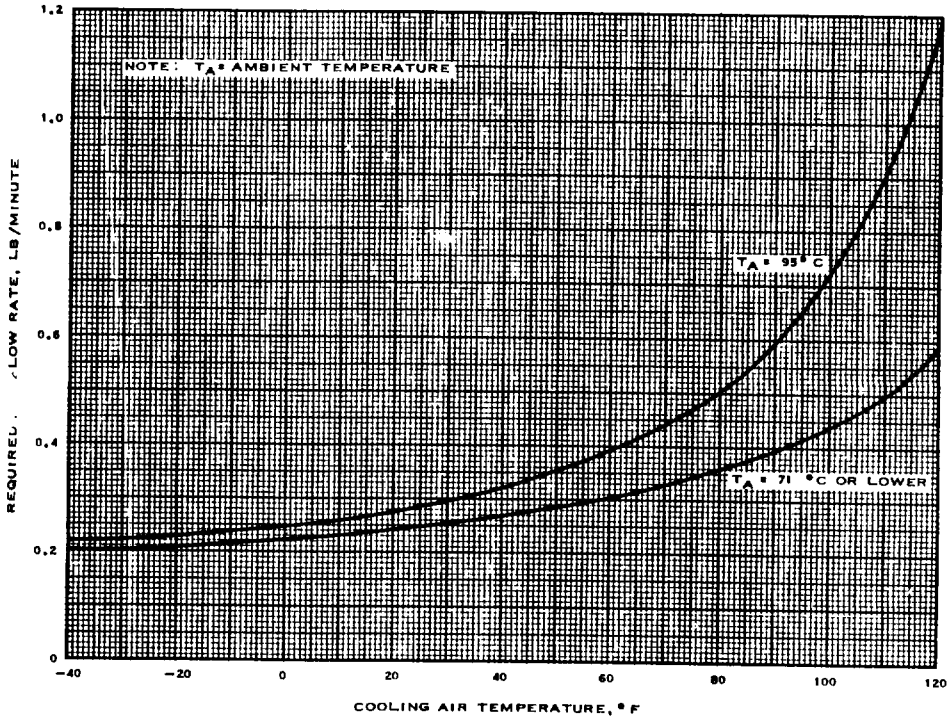


Figure 1. RT-736/ASQ-88 Receiver Transmitter Minimum Cooling Air Requirements

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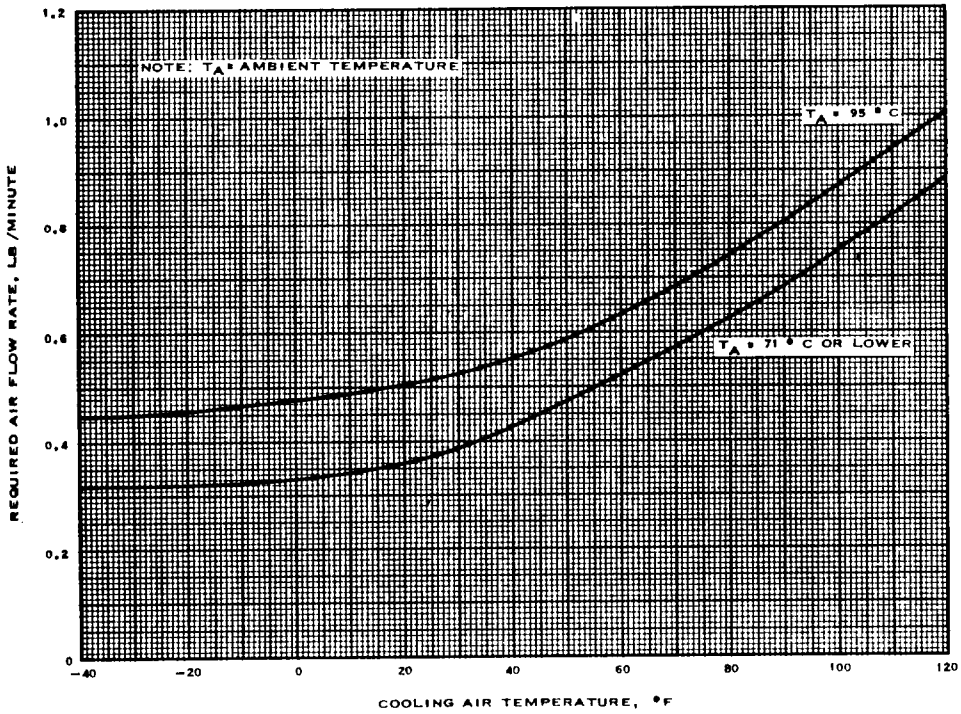


Figure 2. KY-531/ASQ-88 Pulse Decoder Minimum Cooling Air Requirements

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81105(WP) Receiver-Transmitter, Radio RT-736(*)/ASQ-88 and Decoder, Pulse KY-531

ORGANIZATION *(Of submitter)*	CITY AND STATE (*)/ASQ-88

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8279

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

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